# Gamma fn

$n! = n(n-1)!$ for all $x$ except non-positive

$f(x) = x f(x-1)$

$f(x) = \int_0^\infty g(t,x)\,dt$

$\int_0^\infty g(t,x)\,dt = x \cdot \int_0^\infty g(t,x-1)\,dt$

$g(t,x) = x \cdot g(t,x-1)$

$g(t,x) = t^{x-1} e^{-t}$

$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\,dt$

$\Gamma(n+1) = n!$

## Gamma dist

$f(x;\alpha,\lambda) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} = \int_0^x f(u:\alpha,\lambda)\,du = \frac{\gamma(\alpha,\lambda x)}{\Gamma(\alpha)}$

$\mu = \frac{\alpha}{\lambda}$ $\quad var = \frac{\alpha}{\lambda^2}$

* conjugate prior for poisson & exp dist

apps * Model +ve real valued random var
such as time between events in poisson process
duration of a task or process
size of RV

* Reg Models
used as response dist
↓ if
Model $<$ +ve, cont

- Model rela betwn dep variable & ind variables
I follow gamma dist

- using GLM with gamma response dist → rela $<$ dep, indep

* Survival Times (+ve, cont)
- time until patient experience certain event
- prop patient surviving beyond certain pt

## * SVM

crucial hyperparameter (non-linear kernel)

gamma fn defines behaviour of decision boundary

gamma fn value →
- high →
  - influence of each training example min
  - noise in train data
  - overfitting
  - DB very tight
- low →
  - influence of each training example reaches farther
  - less complex DB
  - underfitting

## Hessian Matrix

H: contain second partial der [curvature info]

in NN tell us How to take step

help in regularization (reach l.min)

not used in DL → computational power

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

eigen values →
- $>0$ PD l.min
- $<0$ ND l.max
- mixed Inde saddle pt
- zero singular test inclusive

use its inverse to find critical pt
approximate errors
statistical diagnostics [fisher info matrix in max likelihood est]
CV det of Hessian (DoH) for feature detect